

Jordan Thompson

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WORK EXPERIENCE

VRNDA Labs

June 2026 - Present

Junior Mechatronics Engineer

Richmond, BC

- Engineered fluid delivery systems utilizing NEMA 17 stepper motors, CL42T-V41 drivers, and 3D printed parts designed in Fusion360; achieving 0.1–1200 mL/hr flow rates via microstepping at 51,200 steps/rev.
- Designed a minimum viable prototype for a modular syringe pump; from requirements to a CAD design with a bill of materials in 2 weeks.
- Created drawings in Fusion360 of an end effector hot swapping mechanism based on custom and off the shelf components demonstrating the assembly, dimensioning and tolerances (GD&T).

Kardium Inc

May 2025 – Aug 2025

Biomedical Test Engineer Co-op

Burnaby, BC

- Designed and executed design verification test protocols (TOC and FT-IR) to validate cleaning and sterilization efficacy of intravascular catheter-based ablation devices, generating data that directly supported an FDA regulatory submission.
- Ensured ISO 10993 biocompatibility compliance within an ISO 13485 Quality System; conducted non-parametric statistical analysis in Minitab and participated in change controls, ECOs, and deviations.
- Collaborated with outside vendors to source and qualify components and materials for test fixtures and protocols.

Bioaugmentative Interfaces Lab

Nov 2023 - April 2025

Mechatronics Co-op Student

Vancouver, BC

- Designed and built a phantom bladder rig (3D-printed stand, flow meter, and force sensing) to mechanically validate flow rates and actuation performance of an implantable device ahead of in-vivo animal trials.
- Contributed to the mechanical design and packaging of a proof-of-concept device housing, supporting migration to a miniaturized form factor for rodent implantation.

Mechatronics Research Assistant

Vancouver, BC

- Optimized a shape-memory alloy actuator's duty cycle, achieving 78% liquid voiding efficiency while restricting temperature increases to <2°C.
- Designed and iterated upon modular 3D-printed molds in OnShape to streamline actuator insulation, minimize air bubbles, and improve manufacturing repeatability.

SKILLS

CAD and Rapid Prototyping: SolidWorks, GD&T, Fusion360, 3D Printing (FDM & MSLA), Laser Cutting, Hand Tools

Programming Languages: C/C++, MATLAB, Python

Electronics and PCB: Altium, Eagle, Oscilloscope, Digital Multimeter, Soldering

PROJECTS

Dr. Wobbles, Open Source Pediatric Pain Relief Device - [Website](#)

2025 - 2026

- Designed a screwless, modular 3D-printed housing in OnShape, enabling intuitive non-technical user assembly in under 13 minutes for an open-source medical device that provides pain relief through vibration and cooling.
- Engineered a motor-driving PCB that delivers stable varying vibrational frequency modes (68–172 Hz) with a brushed DC motor, using noise filtering techniques to prevent MCU brownouts (lowpass filter + flyback diode).
- Developed a comprehensive Design History File (DHF) consisting of design requirements, test protocols and drawings.

Spatio-Temporal Electromyography Data Analysis Algorithm

2025 - 2026

- Built an automated TKEO-based detection algorithm to successfully isolate transient signal features in low signal-to-noise environments.
- Validated spatial displacement predictions against 3D-scanned physical coordinates, significantly reducing euclidean distance errors across dynamic tests.

EDUCATION

University of British Columbia BAsC, Biomedical Engineering specialized in Mechatronics, Graduated April 2026